



OpenPipeline Processing

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Configuring Pipeline Stages for Log Transformation

This notebook covers OpenPipeline processing stages: parsing, enrichment, metric extraction, event generation, bucket routing, and filtering.

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Prerequisites

-  Access to a Dynatrace environment with log data
-  OpenPipeline configuration permissions
-  Completed OPLOGS-01 and OPLOGS-02

2. Parsing & Field Extraction

Parsing extracts structured fields from unstructured log content **at ingestion time**.

DPL (Dynatrace Pattern Language) Matchers

DPL Pattern Matchers Quick Reference

Dynatrace Pattern Language for Log Parsing

Text Matchers

LD	Line data (greedy)	"error in module"
WORD	Single word	"ERROR"
DATA	Any text (non-greedy)	"any content"
NSPACE	Non-whitespace	"user@host.com"
SPACE	Whitespace characters	" " (tabs/spaces)

Number Matchers

INT	Integer	-1, 42, 1000
LONG	Long integer	9223372036854775807
DOUBLE	Decimal number	3.14, -0.5
BOOLEAN	true/false	true, false
HEX	Hexadecimal	0x1F, DEADBEEF

Network / Identity

IPADDR	IPv4/IPv6 address	10.0.0.1, ::1
IPV4	IPv4 only	192.168.1.100
IPV6	IPv6 only	2001:db8::1
HTTPURI	HTTP URL path	/api/v1/users
UUID	UUID format	alb2c3d4-...

Time / Date

TIMESTAMP	ISO timestamp	2024-01-15T...
DATE	Date (YYYY-MM-DD)	2024-01-15
TIME	Time (HH:MM:SS)	14:30:00
EPOCH	Unix timestamp	1705329600
DURATION	Time duration	2h30m, 1d

Syntax: MATCHER.fieldname (e.g., INT:status_code) or MATCHER (no capture)

Matcher	Description	Example Match
LD	Line data (to delimiter)	Any text
INT	Integer	42 , -17
DOUBLE	Decimal number	3.14 , -0.5
IPADDR	IP address	192.168.1.1
WORD	Word characters	hello123
TIMESTAMP	Date/time	Various formats
JSON	JSON object/array	{"key": "value"}

OpenPipeline Parse Processor

```
# Parse HTTP access logs
processors:
  - name: parse-http-logs
    type: dql
    source: content
    dql: |
      parse content, "IPADDR:client_ip SPACE LD SPACE '['
TIMESTAMP:request_time ']'
                        SPACE '\"' LD:method SPACE LD:path SPACE LD '\"'
                        SPACE INT:status SPACE INT:bytes"
```

```
// Discover log patterns for parsing design
fetch logs, from: now() - 1h
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Test parse pattern before configuring in OpenPipeline
fetch logs, from: now() - 1h
| filter contains(content, "GET") OR contains(content, "POST")
| parse content, "LD:method SPACE '/' LD:path SPACE INT:status_code"
| filter isNotNull(status_code)
| summarize {count = count()}, by: {method, status_code}
| sort count desc
| limit 15
```

```
// Verify fields already extracted by OpenPipeline
fetch logs, from: now() - 1h
| limit 100
| summarize {
  has_loglevel = countIf(isNotNull(loglevel)),
  has_status = countIf(isNotNull(status)),
  has_trace_id = countIf(isNotNull(trace_id)),
  has_span_id = countIf(isNotNull(span_id))
}
```

3. Metric Extraction from Logs

Extract metrics with dimensions from log data at ingestion time. This is powerful for:

- Creating SLIs from log patterns
- Building dashboards without log queries
- Enabling metric-based alerting

OpenPipeline Metric Extraction

```
# Extract request duration metric from logs
processors:
  - name: extract-request-metric
    type: metric
    enabled: true
    condition: contains(content, "duration=")
    metricKey: log.request.duration
    dimensions:
      - service: k8s.namespace.name
      - method: extracted_method
      - status: extracted_status
    value: extracted_duration_ms
```

Common Metric Extraction Patterns

Log Pattern	Metric	Dimensions
Request completed	log.request.count	service, method, status
Response time	log.response.duration	endpoint, status_code
Error occurred	log.error.count	error_type, service
Queue depth	log.queue.size	queue_name

```
// Identify logs with numeric values for metric extraction
fetch logs, from: now() - 1h
| filter contains(content, "duration")
      OR contains(content, "latency")
      OR contains(content, "time=")
      OR contains(content, "ms")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 15
```

```
// Simulate metric extraction: count by dimensions
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
    request_count = count(),
    error_count = countIf(loglevel == "ERROR")
}, by: {k8s.namespace.name, k8s.workload.name}
| fieldsAdd error_rate = round((error_count * 100.0) / request_count,
decimals: 2)
| sort error_count desc
| limit 15
```

```
// Preview: What dimensions would be valuable?
fetch logs, from: now() - 1h
| summarize {
    unique_namespaces = countDistinct(k8s.namespace.name),
    unique_workloads = countDistinct(k8s.workload.name),
    unique_pods = countDistinct(k8s.pod.name),
    unique_hosts = countDistinct(dt.entity.host),
    unique_sources = countDistinct(dt.openpipeline.source)
}
```

4. Attribute Creation & Enrichment

Add **computed attributes** to logs for enhanced analysis and filtering.

OpenPipeline Field Processor

```
# Add computed attributes
processors:
  - name: enrich-environment
    type: fieldsAdd
    fields:
      - name: environment
        value: |
          if(contains(k8s.namespace.name, "prod"), "production",
            else: if(contains(k8s.namespace.name, "staging"), "staging",
              else: "development"))

      - name: severity_score
        value: |
          if(loglevel == "ERROR", 3,
            else: if(loglevel == "WARN", 2,
              else: 1))

      - name: team_owner
        value: |
          if(contains(k8s.namespace.name, "payment"), "platform-team",
            else: if(contains(k8s.namespace.name, "frontend"), "web-team",
              else: "unknown"))
```

Common Attribute Patterns

Attribute	Source	Use Case
environment	namespace name	Filter prod vs dev
team_owner	namespace/labels	Route alerts
severity_score	loglevel	Prioritization
log_category	content patterns	Classification

```
// Preview attribute creation logic
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd environment = if(contains(k8s.namespace.name, "prod"),
"production",
                        else: if(contains(k8s.namespace.name, "staging"),
"staging",
                        else: "development"))
| fieldsAdd severity_score = if(loglevel == "ERROR", 3,
                        else: if(loglevel == "WARN", 2,
                        else: 1))
| summarize {count = count()}, by: {environment, severity_score, loglevel}
| sort environment asc, severity_score desc
```

```
// Categorize logs by content patterns
fetch logs, from: now() - 1h
| fieldsAdd log_category = if(contains(content, "Exception") OR
contains(content, "Error"), "exception",
                        else: if(contains(content, "request") OR
contains(content, "response"), "http",
                        else: if(contains(content, "database") OR
contains(content, "query"), "database",
                        else: if(contains(content, "auth") OR
contains(content, "login"), "security",
                        else: "general"))))
| summarize {count = count()}, by: {log_category, loglevel}
| sort count desc
```

```
// Identify namespaces for team ownership mapping
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {log_count = count()}, by: {k8s.namespace.name}
| sort log_count desc
| limit 20
```

5. Event Generation from Logs

Create **business events** from specific log patterns. Events flow to Grail and can trigger workflows.

OpenPipeline Event Processor

```
# Generate events from critical log patterns
processors:
  - name: generate-payment-events
    type: bizevents
    enabled: true
    condition: contains(content, "payment") AND contains(content,
"completed")
    eventType: com.example.payment.completed
    attributes:
      - payment_id: extracted_payment_id
      - amount: extracted_amount
      - currency: extracted_currency
      - customer_id: extracted_customer

  - name: generate-error-events
    type: bizevents
    enabled: true
    condition: loglevel == "ERROR" AND contains(content, "critical")
    eventType: com.example.critical.error
    attributes:
      - error_type: extracted_error_type
      - service: k8s.namespace.name
```

Event Use Cases

Log Pattern	Event Type	Purpose
Order completed	order.completed	Business analytics
User signup	user.registered	Funnel tracking
Deployment	deployment.completed	Change tracking
Critical error	error.critical	Incident trigger


```
// Discover patterns suitable for event generation
fetch logs, from: now() - 1h
| filter contains(content, "completed")
      OR contains(content, "success")
      OR contains(content, "failed")
      OR contains(content, "created")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Check existing bizevents (if any generated)
fetch bizevents, from: now() - 24h
| summarize {count = count()}, by: {event.type}
| sort count desc
| limit 15
```

```
// Preview: Logs that would become critical error events
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| filter contains(content, "critical")
      OR contains(content, "fatal")
      OR contains(content, "severe")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| fields timestamp, k8s.namespace.name, content_preview
| sort timestamp desc
| limit 20
```

6. Bucket Routing

Route logs to **appropriate buckets** based on content, source, or computed attributes.

Why Bucket Routing Matters

Bucket	Logs	Retention	Cost Impact
default_logs	Standard	35 days	Baseline
debug_logs	DEBUG/TRACE	7 days	80% savings
audit_logs	Security/compliance	365 days	Compliance
error_logs	Errors only	90 days	Investigation

OpenPipeline Route Processor

```
# Route logs to appropriate buckets
processors:
  - name: route-debug-logs
    type: route
    condition: loglevel == "DEBUG" OR loglevel == "TRACE"
    bucket: debug_logs

  - name: route-audit-logs
    type: route
    condition: contains(content, "audit") OR contains(content, "security")
    bucket: audit_logs

  - name: route-error-logs
    type: route
    condition: loglevel == "ERROR" OR loglevel == "FATAL"
    bucket: error_logs
```

```
// Current bucket distribution
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.system.bucket}
| sort count desc
```

```
// Preview routing decisions
fetch logs, from: now() - 1h
| fieldsAdd target_bucket = if(loglevel == "DEBUG" OR loglevel == "TRACE",
"debug_logs",
                                else: if(loglevel == "ERROR" OR loglevel ==
"FATAL", "error_logs",
                                else: if(contains(content, "audit") OR
contains(content, "security"), "audit_logs",
                                else: "default_logs"))
| summarize {count = count()}, by: {target_bucket, loglevel}
| sort count desc
```

```
// Estimate storage savings from routing
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    total_logs = count(),
    total_mb = sum(content_bytes) / 1048576.0,
    debug_logs = countIf(loglevel == "DEBUG" OR loglevel == "TRACE"),
    debug_mb = sum(if(loglevel == "DEBUG" OR loglevel == "TRACE",
content_bytes, else: 0)) / 1048576.0
}
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd savings_7d_vs_35d = round(debug_mb * 0.8, decimals: 2)
```

7. Filtering & Sampling

Reduce log volume by **filtering** or **sampling** at ingestion.

OpenPipeline Filter Processor

```
# Drop noisy, low-value logs
processors:
  - name: drop-health-checks
    type: filter
    condition: contains(content, "health") AND loglevel == "INFO"
    action: drop

  - name: drop-heartbeats
    type: filter
    condition: contains(content, "heartbeat") OR contains(content,
"keepalive")
    action: drop

  - name: sample-debug-logs
    type: sample
    condition: loglevel == "DEBUG"
    rate: 0.1 # Keep only 10% of DEBUG logs
```

```
// Identify candidates for filtering (high volume, low value)
fetch logs, from: now() - 1h
| filter contains(content, "health")
      OR contains(content, "heartbeat")
      OR contains(content, "alive")
      OR contains(content, "ready")
| summarize {count = count()}, by: {k8s.namespace.name, loglevel}
| sort count desc
| limit 15
```

```
// Calculate potential savings from filtering health checks
fetch logs, from: now() - 24h
| summarize {
  total_logs = count(),
  health_logs = countIf(contains(content, "health") OR contains(content,
"heartbeat")),
  debug_logs = countIf(loglevel == "DEBUG")
}
| fieldsAdd health_pct = round((health_logs * 100.0) / total_logs, decimals:
1)
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd potential_reduction = round(((health_logs + debug_logs * 0.9) *
100.0) / total_logs, decimals: 1)
```

8. Complete Pipeline Example

Here's a comprehensive OpenPipeline configuration:

```
# Complete OpenPipeline configuration for logs
name: production-log-pipeline
enabled: true

processors:
  # 1. FILTER - Remove unwanted logs first
  - name: drop-health-checks
    type: filter
    condition: contains(content, "health") AND loglevel == "INFO"
    action: drop

  # 2. PARSE - Extract structured fields
  - name: parse-http-logs
    type: dql
    condition: contains(content, "HTTP")
    dql: parse content, "LD:method SPACE '/' LD:path SPACE INT:status SPACE
INT:duration_ms"

  # 3. ENRICH - Add computed attributes
  - name: add-environment
    type: fieldsAdd
    fields:
      - name: environment
        value: if(contains(k8s.namespace.name, "prod"), "production", else:
"non-prod")

  # 4. MASK - Protect sensitive data
  - name: mask-emails
    type: mask
    field: content
    pattern: "[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.\.[A-Za-z]{2,}"
    replacement: "[EMAIL-MASKED]"

  # 5. EXTRACT METRICS - Create dimensional metrics
  - name: extract-request-count
    type: metric
    metricKey: log.http.requests
    dimensions:
      - namespace: k8s.namespace.name
      - status: status

  # 6. GENERATE EVENTS - Create business events
  - name: generate-error-events
    type: bizevents
    condition: loglevel == "ERROR"
    eventType: com.app.error
```

```
# 7. ROUTE - Send to appropriate bucket
- name: route-debug
  type: route
  condition: loglevel == "DEBUG"
  bucket: debug_logs
```

```
// Verify current pipeline processing
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    with_pipeline = countIf(isNotNull(dt.openpipeline.pipelines)),
    unique_pipelines = countDistinct(dt.openpipeline.pipelines)
}, by: {dt.openpipeline.source}
| sort total_logs desc
```

```
// Pipeline processing summary
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.openpipeline.pipelines,
dt.system.bucket}
| sort count desc
```



Summary

In this notebook, you learned:

- ✅ **Parsing** - Extract structured fields at ingestion
- ✅ **Metrics** - Create dimensional metrics from logs
- ✅ **Attributes** - Add computed fields for analysis
- ✅ **Events** - Generate business events from patterns
- ✅ **Routing** - Direct logs to appropriate buckets
- ✅ **Filtering** - Drop/sample low-value logs

Key Takeaway

Process at ingestion, not at query time. OpenPipeline processing is fundamental to cost optimization, data quality, and operational efficiency.

Next Steps

Continue to **OPLOGS-04: Buckets & Data Governance** to learn about storage management and retention policies.

References

- [OpenPipeline Overview](#)
 - [OpenPipeline Processors](#)
 - [Metric Extraction](#)
 - [Event Generation](#)
 - [DPL Pattern Language](#)
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